

Business Insights Report (Exploratory Data Analysis – Marketing Campaign Dataset)

Objective:

The objective of this project is to perform **Exploratory Data Analysis (EDA)** on a marketing campaign dataset and understand campaign performance using business insights instead of Machine Learning. The analysis focuses on identifying which marketing channels generate better returns and recommending where future marketing budget should be allocated based on **Return on Investment (ROI)** and overall funnel performance.

Introduction:

Marketing campaigns are used by companies to attract customers and increase sales. However, spending more money on advertising does not always lead to better results. Businesses must analyze campaign performance to understand which channels produce higher revenue and better conversion.

In this project, a marketing campaign dataset was analyzed to study customer engagement and campaign effectiveness. Since no external dataset was provided, a sample dataset was created containing campaign metrics such as cost, revenue, impressions, clicks, and conversions.

The analysis was completed through data cleaning, visualization, and business interpretation.

Dataset Description:

The dataset contains marketing information from different channels.

The columns used are:

- **Channel** – Marketing platform used for campaigns
- **Cost** – Total amount spent on advertising
- **Revenue** – Income generated from campaigns

- **Impressions** – Number of times advertisements were shown
 - **Clicks** – Number of users who clicked
 - **Conversions** – Number of successful customer actions
 - **ROI** – Return on Investment calculated for business analysis
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Data Cleaning and Preparation:

Before performing analysis, the dataset was prepared to improve data quality.

The following cleaning steps were performed:

1. Loaded the dataset into Python using Pandas.
2. Checked dataset structure and data types.
3. Verified missing values.
4. Removed duplicate records if present.
5. Calculated ROI using revenue and campaign cost.
6. Prepared data for visualization.

These steps ensured accurate analysis and reliable business insights.

Exploratory Data Analysis (EDA):

1. Revenue Analysis:

Revenue generated by each marketing channel was compared using bar charts.

Observation:

- Some channels generated significantly higher revenue than others.
- High revenue channels indicate stronger customer engagement and better campaign effectiveness.

Business

Insight:

Revenue alone should not determine budget decisions because some campaigns may require very high spending.

2. ROI (Return on Investment) Analysis:

ROI was calculated to measure campaign profitability.

Formula:

$$\text{ROI} = ((\text{Revenue} - \text{Cost}) \div \text{Cost}) \times 100$$

Observation:

- Certain channels delivered stronger returns with lower investment.
- Higher ROI channels were more efficient than channels with larger spending.

Business

Insight:

ROI is more useful than total revenue because it measures actual business value.

3. Marketing Funnel Analysis:

A funnel analysis was created using:

Impressions → Clicks → Conversions

Observation:

- Large numbers of users viewed advertisements.
- Fewer users clicked.
- Even fewer completed conversion actions.

Business

Insight:

This drop is normal but indicates opportunities to improve targeting, content quality, and customer experience.

Key Findings:

After analyzing the dataset, the following findings were identified:

- Marketing performance differs across channels.
 - Some channels produce better revenue while spending less.
 - Conversion rates decrease at each stage of the marketing funnel.
 - ROI helps identify the most profitable campaigns.
 - Budget allocation should depend on efficiency rather than total spending.
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Recommendations:

Based on the analysis, the following recommendations are suggested:

Increase Budget For:

- Channels with consistently high ROI
- Campaigns showing strong conversion performance
- Cost-efficient marketing activities

Optimize:

- Medium-performing channels through better targeting and content improvements

Reduce Budget For:

- Channels with high spending but low returns
 - Campaigns with weak conversion performance
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Conclusion

This Exploratory Data Analysis successfully evaluated marketing campaign performance and provided business-focused insights. Instead of building a predictive model, the project concentrated on understanding campaign behavior and making practical decisions.

The analysis shows that **ROI-based decision making is more effective than focusing only on total revenue or advertising cost**. By investing more budget into high-performing channels and optimizing weaker campaigns, organizations can improve marketing efficiency and maximize overall business growth.

Tools Used:

- Python
- Pandas
- Matplotlib
- Seaborn
- Google Colab